SAINT MICHAEL COLLEGE OF CARAGA

UTILIZING JACKFRUIT SEED AS AN INGREDIENT IN BAKED PRODUCT:

A HIGAONON-INSPIRED PRODUCT DEVELOPMENT

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CHAPTER I

THE PROBLEM AND ITS SCOPE

1.1 Background of the study

The jackfruit (Artocarpus heterophyllus) has a sweet, fragrant flavor and a slightly crunchy texture. It is commonly used in home-cooked dishes and sweets. As stated by Celestial & Trinidad (2023), jackfruit seeds contain high amounts of protein, dietary fibre, and carbohydrates. Hiba & Geetha Lekshmi (2023) also found that fruit byproducts, such as seeds, constitute 10-35% of the total fruit mass, and are generally discarded following processing, while Alharbi et al. (2021) found that fruit processing industries produce more than 10% of total fruit weight as seed waste each day. This highlights a significant underutilization of a potentially valuable resource. For other fruits, seeds from passion fruit, papaya, and similar fruits are typically discarded (Fonseca et al., 2022; Kawakami et al., 2021). This study will develop healthier baked goods from jackfruit seed waste, inspired by the Higaonon tribe. By utilizing the nutritional value of the seeds, reducing food waste, and integrating Higaonon cultural practices, the study addresses a coherent research problem that combines health, sustainability, and cultural preservation.

Recent studies show that jackfruit seeds can be utilised in beneficial ways. Indian researchers found that about 70% of a jackfruit, mostly the peel and core, is lost during processing, according to Sarangi et al. (2023). Researchers from India and Malaysia found that jackfruit seeds are high in protein, fibre, and antioxidants, making them a healthy flour alternative (Pathak et al., 2022; Suzihaque et al., 2022). Other studies, such as those by Sucharitha Reddy et al. (2022) and Van et al. (2023), also show that jackfruit seed flour can make gluten-free, high-protein, high-fibre, antioxidant-rich foods. Besides this, Juárez-Barrientos et al. (2025) and Kumari et al. (2022) found in Mexico and India that incorporating jackfruit seed flour in extruded snacks increases essential nutrients and protein. Malaysian and Thai studies (Goh et al., 2024; Thatsanasuwan et al., 2023)

show that jackfruit seed flour increases protein, fibre, and mineral content while reducing carbohydrates in baked goods and pasta. Mohammed et al. (2024) stated that jackfruit seed flour is gluten-free but needs careful ingredient balance for good texture. A community study in Byse, India, found its use as food was low, with an average score of 1.77 (Manianga et al., 2024). Although jackfruit seeds have nutritional and economic value, they remain underused, and no Philippine study has created a culturally inspired, nutritious, and sustainable baked product. This gap is relevant in Nasipit, Agusan del Norte, where value-added, culturally based products may support community livelihoods, small farmers' income, and the sustainable use of local resources.

The Philippine Development Plan 2023-2028 supports local communities in creating food products that sustain livelihoods. Rice delicacies in Biliran help residents earn income while preserving cultural heritage (Frances & Barahan, 2023). While they are not popular, local cuisines in Guiguinto, Bulacan, also indicate the identity of the community (Palomares et al., 2023). Studies in the Philippines show jackfruit seeds are nutritious and safe when properly processed, as demonstrated in Pampanga (Angeles City) (Ben & Andalecio, 2021). The sawsawan culture also highlights the cultural significance of even basic condiments (Aster et al., 2023). As stated by Buenavista et al. (2022), the Higaonon people of Bukidnon, in the northern part of Mindanao, utilise 76 edible plants as alternatives to rice. They are known for their traditional knowledge and long-standing use of local resources, making them a suitable model for culturally inspired products. Focusing on the Higaonon tribe highlights traditional knowledge, sustainable practices, and cultural significance. Beyond nutrition, jackfruit seeds hold special meaning as they are used to make coffee, and the taste is believed to reveal a person's emotions. If someone wants to know whether a close friend or family member is upset or distant, they can prepare jackfruit seed coffee. A very bitter taste is interpreted as confirmation. These traditions provide cultural value and guide the development of Higaonon-inspired products. Other regions, such as Lanao del Norte and

Lanao del Sur, maintain food traditions that foster a sense of belonging and cultural identity (Daud, 2025). Even today, Mami delicacies remain an important part of Meranaw celebrations in Marawi, Lanao del Sur, Philippines (Minodar & Bangcola, 2025), and farmers are still able to use traditional pest removal techniques (Valdez, 2023). Nasipit already has a variety of local foods, but no speciality food has been recorded. Recent studies on culturally-inspired baked goods are limited. This research fills this gap by looking at the opportunities of Higaonon-inspired cookies as a food product. The Philippine policy perspective on modernization of agriculture promotes value addition of local produce, under RA 8435 (1997), and recycling of farm waste, e.g., jackfruit seeds, under RA 9003 (2000).

Jackfruit seeds are commonly discarded in Nasipit, Agusan del Norte despite their documented nutritional value and functional properties. Developing prototype cookies using jackfruit seeds will help determine their suitability as a food ingredient based on product quality and consumer acceptability in small-scale production. This study will explore jackfruit seed waste through prototype cookies inspired by Higaonon traditions, supporting sustainable and culturally grounded food products. Many studies show that jackfruit seeds are nutritious but underused locally. This study will provide practical contributions by offering a sustainable, waste-reducing product that can generate income for farmers, households, and small entrepreneurs, while also providing academic contributions for stakeholders such as the College of Tourism and Hospitality Management (CTHM) through research insights, educational use, and product development guidance.

The study will focus only on cookie development and will evaluate their taste, texture, and acceptability among consumers. It will be conducted on a small scale and will exclude large-scale production, marketing, and commercialization.

1.2 Theoretical Framework

The Sustainable Livelihood Framework (SLF) was created by the UK Department for International Development (DFID) during the late 1990s. It states that individuals and groups use what they possess to build a better, more productive life (Fang et al., 2009; Kamariah et al., 2024). Consuming jackfruit seed flour in Higaonon-inspired baked goods will not only yield a novel product but also foster the local economy by providing new livelihood opportunities for farmers, small business owners, and local entrepreneurs in Nasipit. This strategy aligns with the objectives of the Philippine Development Plan 2023-2028, which stimulates community development by developing value-added local products and marketing them.

The Functional Food Theory explains that food can do more than fill our stomachs because it can also help improve health (Palaniappan & Bertozzi, 2016). Jackfruit seed flour is rich in protein, fibre, resistant starch, and antioxidants, which makes it an excellent ingredient for healthier baked products. This idea aligns well with the study's goal of developing cookies that not only provide energy but also promote better health. As more people today look for foods that support their well-being, this product answers that growing demand.

Food and Cultural Identity Theory, developed by Claude Fischler in 1988, describes that food is significant in culture and identity. Preparation, sharing, and enjoying food are all signs of a people's values and traditions. This concept can be substantial to this project, as the objective is to design a baked product inspired by the Higaonon people, in recognition and conservation of their culture. In line with this, Cantarero et al. (2013) found that individuals tend to consume foods associated with their culture, as this helps them develop a stronger sense of identity and belonging. Producing a culturally significant product will also increase local pride, besides bringing something novel and exciting into the market.

1.3 Review of Related Literature

This chapter lists all related literature supporting the researchers' study on "Utilizing Jackfruit Seed As an ingredient in Baked Products: A Higaonon-Inspired Product Development."

Literature of Study

Consumer

The taste of consumers is significant in designing alternative baked goods, especially those that use non-conventional ingredients like a jackfruit seed flour. Some studies have also shown that consumer acceptance of baked goods depends mostly on the sensory attributes such as taste, texture, aroma and appearance. Nordin et al. (2024) tested the application of jackfruit seed flour as a partial substitution of wheat flour in cookies and determined that the formulations with moderate levels of substitution were rated highly satisfactory by consumers, meaning that jackfruit seed flour could be acceptable when the sensory quality was ensured. Likewise, Moolwong et al. (2024) stated that cookies were created with a partial substitution of the wheat flour with jackfruit seed flour, which obtained high overall liking scores, which implies that consumers generally like baked products with the use of this underused ingredient. Nguyen et al. (2023) have corroborated these results by making a comparative analysis of jackfruit seed flour and starch in the baking of cookies and found that sensory properties played a key role in consumer acceptance, and some of the formulations were similar to traditional products. In addition, Aschemann-Witzel et al. (2021) also highlighted that consumers are becoming more accepting of food products made using underutilized or upcycled foods, especially when the products are of good sensory quality and reflect cultural, environmental, or sustainability.

Age

Snacking behavior varies across age groups, with younger consumers generally consuming snacks more frequently than older populations. Studies show that biscuits and cookies are widely

consumed across populations, with many individuals snacking 2–3 times per week, highlighting the prominence of snack foods in daily dietary habits (Abu Seman et al., 2024). Similarly, in urban Thailand, younger consumers (12–18 years) preferred bakery products and potato chips more than older adults, emphasizing age-related differences in snack preferences (Wangcharoen et al., 2004). To further understand these differences, Hymes et al. (2020) analyzed savory snack consumption across two age groups: children aged 2 to 19 years (n = 2,901) and adults aged 20 and older (n = 5,017). The study found that during 2015–2016, 47% of children and 33% of adults consumed savory snacks daily. Similarly, in South Africa, children's consumption of potato chips, corn chips, and cheese curls is notably higher than that of adults (Ibiyemi & Oldewage-Theron, 2023). In the United States, college students reported snacking 2–3 times per day, often choosing less healthy options due to their accessibility (Prapkree et al., 2021). Furthermore, Pineda et al. (2020) found that individuals aged 18–25 are showing growing interest in vegetable snacks and healthier alternatives. In contrast, older adults are generally less open to trying new food products (Bae et al., 2021). When studying children, ethical and safety considerations are essential due to their age and level of understanding. (U.S. Department of Health & Human Services, n.d.). Finally, adolescents aged 12–18 and young adults aged 19–29 consume snacks about 2.9 times per day, with cookies (11.8%), sugar-sweetened beverages (9.4%), and sweets and desserts (9.3%) being the main sources of snack-related energy intake (Tatiane Dos Santos Lopes et al., 2021).

Food Production

The seed flour of jackfruit (JSF) was considered a potential replacement for wheat flour in baked products due to its nutritious nature and other valuable properties. Additionally, the versatility of jackfruit seed flour in baking was emphasized by Brahma & Ray (2023) who demonstrated how it can enhance cookie recipes' sustainability and nutritional value. This applicability corresponds to the growing demand for creative and sustainable food products, which

is driven by the necessity of healthier and permanent meal options (Rabadán et al., 2021). Similarly, Van et al. (2023) confirmed that JSF and jackfruit seed starch might efficiently substitute ordinary wheat flour in cookie formulations at different ratios and measurements, affecting the nutritional quality, physical and chemical properties, and sensory attributes such as taste, aroma, and appearance. Moreover, Moolwong et al. (2023) studied the effect of particle size of JSF on the texture of cookies when wheat flour was partially substituted at 20, 40, and 60 per cent. These research works also help in the potential of associating the application of JSF with recipes of cookies as a new delicacy in Nasipit.

Food Preparation Method

Jackfruit seeds can be prepared and processed in various ways to create nutritious and versatile food ingredients. They are usually also roasted, boiled, fried, or steamed and then utilized in food (Suzihaque et al., 2022). In more specific methods of preparation Priyadharshini (2025), the methods of washing the seeds, peeling off the white aril and brown skin, cutting up the kernels, oven-drying them in 60-80 0 C baths in 10 minutes, and grinding them to a fine powder are detailed. Similarly, Nopa et al. (2025) used washing jackfruit seeds, gelatinizing them in boiling water (4min), chopping them into thin slices, drying at 70 °C (5hours), peeling, grinding, and sieving using a 100-mesh sieve to obtain a gluten-free fettuccine formulation using the flour—Preparation of the Jackfruit seed flour either by the soaking process or the heat treatment. Among The treatments, the water-soaked samples were found to be better in sensory qualities (Ray et al., 2021). Flours obtained (BLJSF, ROJSF, and PCJSF) were used in the preparation of low-fat functional Breadsticks.

Nutritional Content

Nutrition encompasses the essential macro- and micronutrients required for biological activity and optimal health (Singh et al., 2023). Notably, Jackfruit is one of the world's largest edible

fruits that is native to India's Western Ghats and other parts of Asia (Reddy et al., 2022; Gupta et al., 2023). Furthermore, this fruit has an exceptional nutritional value, containing high levels of carbohydrates, proteins, vitamins, minerals, and phytochemicals, including phenolics, flavonoids, and tannins (Kalse & Swami, 2022; Chikkanna, 2021). In addition, Jackfruit demonstrates significant therapeutic properties, exhibiting antioxidant, anti-inflammatory, antimicrobial, antiviral, anticancer, and anti-diabetic activities (Gupta et al., 2022). Moreover, the fruit is also a popular meat substitute due to its unique texture and can be processed into other various commercial products (Gupta et al., 2023; Nansereko & Muyonga, 2021).

Health Benefits

Jackfruit seems to have a lot of antioxidant power, and that might give us several health perks. For instance, Gupta et al. (2023) describes the fruit as having several health-boosting effects, including immune, anti-diabetic, and antioxidant actions. Similarly, Cheng et al. (2025) found the antioxidant effect lines up closely with the fruit's phenolic makeup, which backs up Gupta's idea about the chemistry behind the health boost. In addition, Dhani et al. (2025) Jackfruit compounds have anti-inflammatory, anticancer, and antioxidant properties and improve immune function. Beyond those antioxidant claims, Ye et al. (2025) studied the plant chemicals and found possible anti-obesity, anti-aging, and neuroprotective effects. Moreover, Sudipta Saha et al. (2022) found that jackfruit's nutrients help the heart, blood sugar, and immune system. Finally, (Afotey et al., 2024) showed the fruit might fight malnutrition and be a cheap source of nutrients, turning the lab findings into real-world diet benefits.

Allergens

Oral allergy to jackfruit, linked with cross-reactive plant protein allergen Bet V1, has been associated with latex and birch pollen-related food allergy, but anaphylaxis to jackfruit is rare. For instance birch pollen allergy, affecting 8-16% of the general population, is primarily mediated by

the major allergen Bet v 1, which causes cross-reactive responses with homologous proteins in various foods (Li et al., 2022). Allergy to a heterophyllus is rarely reported and has not been identified as an allergen previously. All patients experienced one or more episodes of anaphylaxis to a heterophyllus. Anaphylaxis is a rare allergic reaction that has been documented in association with latex allergy. In one case, Jalil et al. (2021) described a 21-year-old young man had an anaphylactic response following the consumption of dried jackfruit. He had allergic rhinitis and a latex allergy. He was born too early, and as an infant, he needed neonatal intensive care and numerous surgeries, which may be the root cause of his latex sensitization. Jackfruit and latex were found positive in the skin prick test (Jalil et al., 2021). Overall, rubber latex allergy poses serious risks to healthcare workers, including anaphylaxis.

Jackfruit Integration

Jackfruit seed flour (JSF) has found wide application in bakery products and snacks with the partial replacement of wheat flour, because of its functional and nutritional properties. For instance, Moolwong et al. (2023) able to reach acceptable organoleptical attributes of cookies added with 20% JSF, especially for products made from flour with a particle size between 0.5 and 0.8 mm of JS. Similarly, Minh (2022) shows that bread with sleeping JSF has improved water absorption and emulsion stability without compromising sensory preferences. In addition, Nordin et al. (2024) Found that the cookie with up to 25% replacement of JSF had acceptable taste, appearance, and shelf life. Furthermore, Van et al. (2023) figured out that swapping out 30% of JSF is just right for keeping things nutritious, chemically stable, and tasty. Moreover Reddy et al. (2022) gave us the lowdown on how JSF works in baked items, while Palamthodi et al. (2021) proved that replacing JSF with flakes in food for children who are not eating adequately makes the item healthier, tastier, and more palatable to eat. Finally Pratiwi et al. (2024) Took it up a notch further and showed that replacing JSF.

Cookies Alternative

Cookies are a widely consumed snack across different age groups, making them an ideal vehicle for introducing functional and sustainable ingredients like jackfruit seed flour (JSF). JSF is highly nutritious, containing significant protein and starch (80–85% dry basis), which can enhance the nutritional and functional properties of bakery products (Kushwaha et al., 2021; Kalse & Swami, 2022; Mohammed et al., 2024). Research shows that partial substitution of wheat flour with JSF (25–30%) in cookies improves nutritional composition without compromising sensory quality, particularly at lower substitution levels (Maskey et al., 2020; Moolwong et al., 2023). Similarly, consumer acceptability studies confirm that cookies made with JSF are preferred across diverse age groups, highlighting their potential for the snack market (Nordin et al., 2024). Overall, cookies are suitable for JSF incorporation because they align with both age-related consumption patterns and growing market demand for healthy, functional, and sustainable snacks, providing a practical route for commercial application.

AI Trends in Incorporating Jackfruit Seeds

Future directions emphasize developing AI models, energy-efficient computing, and integrating physics models with AI for enhanced environmental sustainability. For instance, AI techniques, including machine learning, computer vision, and Analytics of data, enable farmers to make informed decisions and optimize resource allocation Victoire (2023). Similarly, zero-waste biological approaches offer a sustainable solution for environmental protection while creating economic value from agricultural waste (Brahma & Ray, 2023). These studies demonstrates AI can optimize resource utilization, enhance productivity, and promote sustainability through technologies like crop health monitoring and predictive analytics. In addition, the integration of AI with Internet of Things (IoT) technologies has improved data collection capabilities through advanced predictive modeling and real-time analytics, particularly in applications such as water

quality and climate monitoring (Sharukh Pasha Mohammed et al., 2025; Miller et al., 2025). Moreover, AI uses remote sensing and data fusion strategies to achieve good accuracy in environmental assessment, with classification accuracy reaching 98.57% for land use changes (Himeur et al., 2022).

Appearance

The way food looks greatly affects what consumers prefer and indicates its quality. For instance, Saleh and Lee (2023) emphasized the need to evaluate food traits like appearance, taste, and texture. They pointed out that visual appeal is just as crucial as other sensory features. Similarly, Moolwong et al. (2023), which showed that cookies with 20% jackfruit seed flour (JSF) scored highest in appearance, even though they were darker and smaller than the control cookies. In addition, Van et al. (2023) discovered that adding more JSF resulted in reduced lightness (L*) and spread ratio in cookies, causing noticeable changes in their visual look. Likewise, Arias et al. (2022) found that Pan de Sal with higher amounts of JSF shifted from golden-brown to darker brown, directly affecting acceptability scores. Supporting these findings, Terigu et al. (2023) showed that mixing JSF with purple sweet potato pulp changed the surface color and overall look of steamed brownies. Moreover, M. et al. (2022) pointed out that cookies with moderate amounts of JSF received good scores for appearance, along with other sensory qualities. In line with this, Palamthodi et al. (2021) confirmed that biscuits with up to 20% JSF kept an acceptable appearance for both the crust and crumb. Finally Maskey et al. (2020) found that small amounts of JSF (about 12.5%) kept cookies looking like wheat ones, but higher amounts made them darker.

Aroma

Adding Jackfruit seed flour to cookies enhances taste and scent, improving the sensory experience. For instance, Girona-Ruíz et al. (2021) Emphasize the role of aroma in sensory marketing, explaining that it helps create an attractive food experience for consumers. Similarly,

Calín-Sánchez and Carbonell-Barrachina (2021) highlighted the aroma is shaped by different compounds, influenced by climate, soil, and processing methods, which adds complexity to food scents. In addition, Rocha et al. (2022), the role of aroma in food quality is challenging to evaluate due to individual preferences, a point also raised by Spada et al. (2020). Moreover, Hiba (2023) found that cookies with 5% aromatic JSF had high sensory scores for aroma, suggesting that it improved the product's scent. Furthermore, Spada et al. (2022) showed that fermented JSF could be added entirely or in part to food formulations to enhance the aroma of chocolate. Overall, these results imply that JSF improves cookies' sensory appeal and adds to their nutritional value with its unique aroma.

Flavor

Researchers concur that its contribution to its flavor is dependent on its processing and the quantity consumed. For instance, Mohammed et al. (2024) reported that jackfruit seeds lack a natural taste, so they are hard to process, and fermentation is capable of enhancing the aroma of baked treats like muffins. Similarly, Suzihaque et al. (2022) also found that jackfruit seed flour positively influences the flavor profile of biscuits and cakes, but in excess, it damages sensory qualities. The other literature emphasizes the change in flavor by the use of the preparation methods. In addition, Teo et al. (2024), jackfruit seeds themselves are sweet in flavor, and when roasted, their flavor is increased as well as their antioxidant ability due to the Maillard reaction. Moreover Mudassir Ur Rahim et al. (2024) highlighted the historical applications of jackfruit seeds in Asian cuisine, in which they are typically cooked or made into desserts, indicating that its flour is a flexible starch source whose taste is acceptable upon cooking.

Taste

The jackfruit seeds are typically eaten in a roasted or cooked form, as a potato alternative, which implies that they have a starchy or relatively neutral flavor (Siddiqui Mudassir Ur Rahim

2024). However, some cultivars with natural levels of sweetness, e.g., J33 (Tekam Yellow), which is around 2730, as well as 2730 respectively (Suzihaque et al., 2022). Similarly, Van et al. (2023) also found in their product applications that cookies made with jackfruit seeds were comparable in taste to regular cookies made with wheat flour and gained a high consumer acceptance. In addition, Priyadharshini (2025) discovered that the 50 per cent substitution of jackfruit seed flour had the highest taste rating (mean 8.6), but increasing the substitution further added a minor element of bitterness and reduced the taste acceptability.

Texture

Jackfruit seeds are inherently complex and rather indigestible, but they get more useful when baked or turned into flour (Siddiqui Mudassir Ur Rahim, 2024). For example, the low bulk density of Jackfruit seed flour typically results in creating a light and porous texture of the product because the granules of starch are driven during milling, and the flour is less compact (Sinha et al., 2025). However, its implementation in products tends to make them more complex, denser, and less expensive (Gowda et al., 2021; Kumari et al., 2022; Mohammed et al., 2024; Ulyarti & Capriola, 2022). Furthermore, they are explained by the presence of high fibre and starch content of jackfruit seed flour (Kumari et al., 2022; Mohammed et al., 2024; Ulyarti & Capriola, 2022). In contrast, Galagaran & Camarin, (2025) established that jackfruit seed cereal and especially formulation B were highly acceptable due to their smooth, balanced texture, which provides a suitable degree of crunch or softness.

Sensory Qualities

Research has explored the effects of incorporating jackfruit seed flour into baked products. For instance, Mohammed et al. (2024) noted that higher incorporation of jackfruit seed flour was negatively affecting the texture, colour, appearance, and overall acceptability of the product. Similarly, Ulyarti & Capriola (2022) also reported that the products with higher substitution levels

were more complex, while Nurul Ummu Habibah (2021) found that the colour of the products was darker. Nevertheless, studies confirm that JSF can be partially substituted for wheat flour without significant loss of quality. Substitution levels of up to 20% are beneficial in terms of color, aroma, taste, and texture (Nirmalawaty et al., 2023; Palamthodi et al., 2021). Moreover, roasting jackfruit seeds has been shown to improve flavor, aroma, and overall sensory appeal (Ngwere & Mongi, 2021; Van et al., 2023). Overall, a substitution of around 30% JSF is recommended to balance nutritional value and sensory quality in baked products (Van et al., 2023).

Shelf Life

The seed of fresh jackfruits typically possesses a shelf life of between 30 days when kept at cool and humid storage facilities (Khan et al., 2021). To extend shelf life, it recommended that the seeds be turned into flour to increase their shelf life (Borgis & Bharati, 2020; Sinha et al., 2025) noted that the large-scale loss of moisture during the formation of the flour is crucial to the extension of the shelf life and the inhibition of microbial decay. Similarly, Suzihaque et al. (2022) observed that jackfruit seed flour is highly stable to storage and maintains its active properties during the average shelf life of the starch-containing food materials. In a more specific study, Priyadharshini (2025) it was discovered that low moisture flour (3.79- 4.98 g) could be held in high density polyethylene (HDPE) at 50 °C and 75% relative humidity (RH) during up to four month.

Impact of Acceptability

Jackfruit (Artocarpus heterophyllus Lam.) seeds are a valuable, underutilized byproduct with significant potential for food applications. For instance, Jackfruit seed flour (JSF) enhances nutrition and improves functional traits like bulk density and water-holding capacity in food products. However, high JSF levels can worsen texture and color due to the lack of gluten, with consumers preferring extruded products over baked ones (Mohammed et al., 2024). In support of this, consumer acceptance studies indicate that higher overall acceptability for extruded products

containing Jackfruit Seed compared to bakery items with similar substitutions (Mohammed et al., 2024). Beyond food applications, Jackfruit Seed shows versatility as a flour for red pigment production for textile colorants and pharmaceutical tablet preparation (Kalse & Swami, 2022). Overall, utilizing these byproducts can improve processing economics while reducing environmental impact (Kalse & Swami, 2022).

Market Potential

Jackfruit (Artocarpus heterophyllus Lam.) has good market potential as a food supplement due to its exceptional nutritional profile and diverse applications. The fruit contains a variety of bioactive compounds, including macro- and micronutrients, and demonstrates multiple health benefits, such as anti-inflammatory, anti-diabetic, antioxidant, and anticancer properties. Similarly, Jackfruit seeds, typically discarded as byproducts, contain 80-85% starch and offer commercial potential as alternative starch sources for food industry applications (Kushwaha et al., 2021). In addition, the fruit's unique texture makes it valuable as a meat substitute, expanding market opportunities (Gupta et al., 2023; Ye et al., 2025). Despite these advantages, jackfruit remains underutilized on a marketable scale in production areas. Current applications include food products like sauces, ice creams, jams, and bakery items, indicating substantial untapped commercial potential (Chikkanna., 2021; Gupta et al., 2023).

Literature of Method

Sun-Drying Jackfruit Seed

One of the most basic processes in dehydrating treatment for jackfruit seed preservation is sun drying, as it is cheaper and applicable in tropical areas. It was reported that jackfruit seeds subjected to heat (35 to 40 °C) inhibited moisture content (7882), thereby stopping the proliferation of microbes, while preserving good protein (14 to 19) and carbohydrate (7783) levels Kamal et al. (2024).The study by Suli Trejo Rodríguez et al. (2021) found that controlled drying of

jackfruit seeds altered the quantity of polyphenols and the seeds' ability to resist free radicals. In this regard, the advantages of Sun-drying are evident, yet it is inferior to emerging methods in terms of preserving all nutrients. A study conducted by Tamsel Ahammed & Sabrina Zaman (2023) on the influence of various drying methods on the chemical composition of jackfruit seed extract highlighted the direct impact of drying method choice on the seeds' nutritional value. To optimize drying processes while maintaining nutritional value, Watanabe et al. (2023) mathematically evaluated jackfruit seed drying, focusing on absorption and reduction.

Toasting Jackfruit Seed

New studies indicate that toasting or roasting jackfruit seeds enhances their nutritional, functional, and sensory properties. Teo et al. (2024) found out that the roasting of jackfruit seed flour increased the water-absorbing properties of flour, as well as the foam, indicating that it could be incorporated into numerous food recipes. To support these conclusions, an in-depth analysis of jackfruit seeds by Dhani et al. (2025) found that appropriate processing, including roasting, can enhance jackfruit seed functionality and health benefits. The potential of jackfruit seed flour for processing and bakery applications was examined by Mohammed et al. (2024) who demonstrated that roasted jackfruit seed flour exhibits favorable bulk density, swelling power, and water and oil absorption capacity. It was also confirmed by Dangang Bossi et al. (2024) that roasting increases the nutritional value of food by reducing the levels of anti-nutrients and increasing the levels of phenols and flavonoids.

Blending

Recent studies have indicated that jackfruit seed flour (JSF) can be blended with other flours to enrich the nutritional, functional, and sensory attributes of food products. According to Arivuchudar & Parameshwari (2025) mixing JSF with other ingredients was reported to reduce anti-nutrients and increase phenolic and flavonoid content, thereby enhancing functional

properties. As noted by Teo et al. (2024), the water absorption capacity and foam stability, both crucial parameters in the food industry, are significantly influenced by the mixing of roasted or raw JSF. As supported by Fabil et al. (2024), the blending of JSF is crucial for marketing its natural functional and nutritional attributes across different food formulations. The article (He & Kong, 2025) introduced the unique properties of jackfruit seed starch and noted that its combination with other substances could enhance its processing. In the study, (Juárez-Barrientos et al., 2025) demonstrated the reinforcement of antioxidant capacity and functional properties of extruded snack products using a mixture of JSF and rice and soy flours.

Oven

Baking conditions for cookies prepared with jackfruit seed flour have also been shown to determine texture, appearance, and taste. One study compared using flour to using only starch, and the outcomes were positive, though some flour was used in place of all other ingredients Van et al. (2023) Further evidence indicates that time and temperature regulation during baking are beneficial for preserving quality during wheat flour replacement Nordin et al. (2024) Moreover, Moolwong et al. (2023) and others published recipes demonstrating that standard baking procedures could be used with jackfruit seed flour in small to medium amounts and that the results pleased customers. In Teo et al. (2024) the impact of baking with unroasted, processed, and cooked jackfruit seed flour was examined, and it was found that liquid absorption and texture changed with the intensity of heat, which in turn affected cookie behavior in terms of shape, texture, and mouth feel. These associated studies show that the quality of cookies and other biscuits depends on baking conditions, including temperature, duration, and the ingredient mix before baking.

Packaging

In the marketing, protection, and preservation of food products, packaging plays an important role. It brings together both arts and science to attract consumer attention without compromising product safety Velankar (2024). In the same way, according to Fernqvist et al. (2025) the packaging plays a crucial role in preserving food. The ability to block UV rays and act as a moisture barrier has led to chitosan-based films being identified as a promising, environmentally friendly packaging option due to their biodegradability. Food packaging is essential for preserving product quality, ensuring safety, and extending shelf life Kovačevi´c et al. (2022). Recent developments in nanotechnology have further enhanced these capabilities.

Time

Jackfruit is widely grown in India and other countries bordering it, as well as in parts of Africa. The seeds containing a good amount of carbohydrate and protein after consumption of the edible part of the ripe fruit are mainly disposed of as waste. Wainaina et al. (2021) examined variables that influence cooking time in common beans, showing that cooking time is influenced by complex factors that require careful attention to processing conditions to achieve good, yet nutritionally sound, taste. In line with this, Diaz et al. (2021) conducted a review of ovenized MEMS gyroscopes and highlighted temperature-control systems that employ closed-loop processes to maintain a constant temperature and prevent performance changes. Supported by (Ceviz et al., 2024), who offered an in-depth review of solar cooking systems, their design, functionality, and practical use, and found the obstacles to the wide use of solar cooking.

Temperature

Temperature influences food in various ways, including preparation, storage, consumption, and safety. In cooking, temperature controls important changes in proteins, the gelatinisation of starch, and the formation of Maillard products, which result in textural and taste changes Pereira

(2025). In storage, too, temperature plays an important role in food, as temperatures above the optimum help hasten the formation of ammonia, lipid oxidation, loss of amino acids, and changes in the texture of long-life foods Šopík et al. (2022). Different processing techniques can impart different properties. For example, K Joy et al. (2022) found that cold plasma treatment increased the water solubility of food materials from 9.65 g/g to 14.11 g/g and enhanced hydration capacity. Serving temperature also influences physiological responses. It is found that hot meals lead to a greater liberation of satiating hormones GLP-1 and CCK, than cold meals, which seem to increase satiety irrespective of saturating matter Hamid et al. (2024). The climate changes do produce the complex emerging problems of food safety due to temperature effects that occur to cold chain, increased microbial growth and toxicity, increased chemical contaminants, disruption of agricultural systems, which will lead to a reduction in food security and food safety globally (Duchenne-Moutien & Neetoo, 2021).

Butter

Butter derived from whey cream and milk cream possesses comparable chemical parameters, colourimetric values, and fatty acid compositions, indicating the predominance of palmitic and myristic acids Aydemir & Altun (2024). However, whey butter generally has lower fat and higher moisture, NaCl, and some minerals, while milk butter contains higher levels of protein, carotenoids, and tocopherols Aydemir & Altun (2024). Material ratios are of particular importance: sucrose, butter, and egg liquid have positive coefficients of correlation with the production of AGEs, with the presence of sucrose particularly favouring the production of HMF and 3-deoxyglucosone Hu et al. (2022). Storage conditions alter water activity and gloss in butters with increasing storage temperatures, resulting in increased gloss and changes in colour indices; whey butter has higher cholesterol but initially lower levels of cholesterol oxidation products, which subsequently increase with time (Cais-Sokolińska et al., 2024).

Granulated Sugar

Value-added functions of granulated sugar beyond sweetening include contributing to structure, shelf life, and moisture retention in a baked product, delaying the gelatinisation of starch, and contributing to colour and flavour (Dana & Sonia, 2024; Huțu & Amariei, 2024). However, high refined sugar consumption is related to non-communicable diseases, dental caries, overweight, and neurodevelopmental disorders in children Arshad et al. (2022). Sugar was known as one of the most common ingredient groups associated with an increased risk of death or disease in dessert recipes Wallach et al. (2023). For healthy recipes, sugar can be substituted at levels of 10-100% with functional ingredients. Among the most successful substitutes from technological and consumer perception perspectives are apple puree, inulin, oligofructose, stevia, apple pomace, polydextrose, dried apples, and date powder/syrup (Dana & Sonia, 2024; Huțu & Amariei, 2024). For instance, sugar sweetness perception could be augmented by modifying its particle size to reduce its content Richardson et al. (2021). These replacements may improve texture, sensory quality, nutrition, and consumer appeal, while retaining essential sugar functions for baking.

Brown Sugar

Aside from its sweetening abilities, brown sugar has considerable potential. Brown sugar has medium to high phenolic class content. In the case of refined sugars, polyphenols undergo degradation Khoo et al. (2023). In particular, brown sugars are rich in polyphenols, such as hydroxybenzoic acids (protocatechuic, p-hydroxybenzoic, vanillic, and syringic) and hydroxycinnamic acids (chlorogenic, p-coumaric, and ferulic). In addition, chlorogenic acid (29.24%) and syringic acid (31.31%) are the most abundant Wang et al. (2023). Some brown sugar sweeteners, as in the case of those from dark Barbados molasses, proportionately contain higher phenolic content compared to many unrefined, direct-from-sap sweeteners. Azlan et al. (2023). Brown sugar polyphenols possess strong anti-hyperglycemic properties due to their potent

inhibition of α-amylase and α-glucosidase enzymes Wang et al. (2023). Natural sweeteners generally offer high sweetness, low-calorie content, good solubility, high stability, and minimal toxic side effects compared to synthetic alternatives Sun et al. (2024). The production of natural sweeteners in Indonesia appears promising, with 15 potential plants identified across various areas, mainly producing saccharide-type sweeteners from cassava, corn, and coconut (Islamy, 2024).

Eggs

Eggs are crucial in breads, especially gluten-free ones. The egg yolk contains fat-soluble vitamins A, D, E, and K, along with minerals, which contribute to the nutritional value and functional characteristics that help bakery products gel and separate into different phases Patil et al. (2025). The research by Yang et al. (2022) demonstrated that egg white protein particles enhance the appearance, texture, and mouthfeel of cookies by stabilising emulsion droplets. Egg whites also seem to perform better than egg yolk in gluten-free bread production and result in higher bread volumes (2.60 cm^3/g vs 2.07 cm^3/g), more stable cell structures, better moisture retention ability, as well as superior springiness of the crumb (Ahlborn et al. (2022). The egg protein helped maintain gas cells in the gluten-free system, and CO2 support during baking was found to be challenging Waziiroh et al. (2023). In gluten-free bread, egg albumin shows good functionality, and, among other factors, protein solubility is important, as foaming/forming and mixing properties are essential for better bread quality (Waziiroh et al., 2023).

Vanilla Extract

Vanilla (Vanilla planifolia) extract is used to add flavour to baked goods, giving them a sweet, aromatic taste that makes them more palatable overall and helps reduce glucose dependency. It provides magnesium and potassium, which support kidney, heart, and muscle function, and has antidepressant and antimicrobial benefits. In the study of Buitimea-Cantúa et al. (2022)

demonstrated that high hydrostatic pressure (HHP) can improve vanillin content by up to 138% and increase enzyme activity by up to 400%, attributed to tissue decompartmentaliation and enzymatic conformational changes. The vanilla extract increased the perception of sweetness by aroma–taste interaction with sugar, which could lead to the reduction of sugar consumption to the extent of (15%) in water and (13%) in coconut-milk desserts, while the acceptance with the consumer is not negatively affected (Soontrunnarudrungsri et al., 2023)

All-purpose flour

Jackfruit seed flour can be a good alternative to wheat flour, as it is rich in nutrients. According to Siddiqui Mudassir Ur Rahim (2024). JSF has a high protein, fibre, and resistant starch content, along with a low moisture content, which enhances the nutritional value and shelf life of baked products. According to Mohammed et al. (2024) substituting wheat flour at a level between 20-30 % with JSF (jackfruit seed flour) improves its protein, fibre, and mineral content. However, beyond this level, excess incorporation can adversely affect texture, volume, and colour due to poor gluten development. Van et al. (2023) made cookies with 30% JSF. According to their research, cookies made with JSF have better fibre and polyphenol content. As a result, the cookies required a higher water addition to regain the desired spreadability and texture. Pizza dough can incorporate 30% JSF without affecting sensory quality Nompo et al. (2024). Wafer cones with 60% JSF showed the highest crispness and acceptability, and 80% incorporation increased protein content by about 14.5% according to (Kushwaha et al., 2021).

Baking Soda

Sodium bicarbonate (baking soda) is a leavening agent that releases carbon dioxide upon mixing with acidic ingredients, creating a lighter crumb structure and enhancing the texture of baked goods. Asamoah et al. (2023) demonstrated that increasing baking powder levels significantly increases batter specific volume and porosity, though with diminishing returns as

levels approach maximum concentrations. It also acts as a pH regulator in dough: lower amounts yield a better white crumb, while higher amounts raise the pH, resulting in a darker crust. A satisfactory addition of approximately 2% of the flour weight to the total formula tends to improve dough stability, yielding greater viscosity and aeration. In contrast, larger additions (approximately 3%) cause coarser pores and over-browning Constantinescu et al. (2022). In addition, bread leavened with soda reduces the phytic acid content of flat wheat breads by 22-51%, resulting in greater solubility and increased mineral damage (Ghaffar, 2025).

Salt

In bakery products, Salt (NaCl) is necessary for stabilising gluten, controlling yeast fermentation, and maintaining dough handling and crumb structure. These are incremental stealth cuts, salt replacers, and flavour boosters to keep the products tasting good Ferrari et al. (2022). Research has investigated the use of fermented brown macroalgae (e.g., Saccharina latissima) as partial salt substitutes to lower sodium intake whilst maintaining flavour and texture. They provide umami flavouring that can compensate for lower sodium levels while maintaining overall dough handling and bread quality. They are rich in minerals and contain free glutamate Tyl et al. (2024). Reducing sodium chloride intake to 0.6% or lower reduces the risk of cerebrovascular accidents by 23% Pashaei et al. (2021). It will tend to increase the nutrient values of the bread due to the additional potassium, calcium, and magnesium contributed by these salts, which tend to lower the blood pressure, to a large extent, due to improvement in bone health and metabolism of nutrients, respectively (Codină et al., 2021; Pashaei et al., 2021).

Chocolate Chips

The primary role of chocolate chips is to support flavour in cookie formulas, and chocolate intensity is the best predictor of consumer liking and purchase intent. A lack of flavour reduces the acceptability of cookies, especially in gluten-free cookies Frauenhoffer et al. (2025). Moreover,

protein-enrichment (i.e., the addition of cricket powder) and high digestibility are reformulations proving that chocolate-chip protein-enriched cookies may be (approximately 60%) highly digestible, and retain better emulsifying properties, but the further the introduction of alternative ingredients, the darker the cookies and the softer the consistency Aleman et al. (2022) found that cocoa bran has a high fibre content (≈ 50%) and phenolic compounds that enhance water-holding capacity, reduce mass loss, and reduce spread, producing baked products with adequate structure and diameter. (Molu et al., 2021) prepared jackfruit-seed chocolate with 30 % roasted jackfruit seed flour and 50 % cocoa butter, which provided good overall nutritional value and sensorial acceptability (8.31).

Lemon Zest

Because of its dietary fibre, vitamin C, essential oils (limonene), and minerals, lemon zest is a healthy ingredient in baked goods that provides antioxidant, antibacterial, and anti-inflammatory effects Patil et al. (2025). Lemon peel extracts are abundant sources of flavonoids, including eriocitrin and hesperidin, which exhibit radical-scavenging and anti-diabetic activities. The process of microencapsulation helps retain flavonoids during baking while also reducing bitterness Dellapina et al. (2025). Besides, the researchers claim that incorporating lemon waste into muffins and cookies will not only enhance the fibre, protein, and phenolic content, but also the aroma and acceptability, though a slight bitterness may be perceived at higher levels Mateus et al. (2025). All of these attributes illustrate that lemon zest adds value to functional baked goods, both nutritionally and sensorially (Prajapati et al., 2025).

Milk

Milk is a healthy base for baked goods because it has a lot of calcium and vitamins A, D, and B-complex. During heating, the Maillard reaction and the fat content (2.5–4%) of the fermented milk baked goods together contributed to imparting taste, smell, and color, while

nutrients remained present. The primary milk constituents, casein and whey, are mainly responsible for the high nutritional and functional quality of milk. Lohinova et al. (2025). Milk contains around 80% casein and 20% whey protein. A few Maillard reaction products (MRPs) have shown antimicrobial activity, which can enhance food preservation when used El Hosry et al. (2025). However, excessive Maillard activity can lead to negative consequences, including HMF formation and more difficult protein digestion Qi et al. (2025). Adding calcium to baked products is one of the ways milk improves their overall quality, besides the changes in taste and color caused by the Maillard process. However, color alone should not be the sole criterion for judging product quality; a colorimeter (Lohinova et al., 2025) should be used to obtain a precise measurement.

1.4 Objectives of the Study:

This study will aim to explore the potential of utilizing jackfruit seed waste in healthier baked products, creating a nutritious, plant-based alternative inspired by Higaonon traditions. The specific objectives of the study are as follows:

1. To develop baked goods that make use of jackfruit seed waste, highlight the capabilities of jackfruit seeds as a local, sustainable ingredient.

2. To evaluate the sensory qualities (taste, smell, and texture) of jackfruit seed waste-based baked products, focusing on how the seed waste affects their flavour, texture, aroma, and appearance.

3. To determine how well the baked products made of waste jackfruit seed are accepted by the consumer based on gathering feedback on how it tastes, feels, and how satisfied they are with it, with special focus on the novel application of jackfruit seed waste.

4. To find out the best formulation and concentration of jackfruit seed waste that has a positive effect on nutritional value and still has an acceptable texture, which would give a pleasant experience of eating.

5. To present feasible suggestions regarding the designing, manufacturing, and advertising of jackfruit seed waste-based baked products as healthier substitutes to the traditional baked products.

1.5 Definition of Terms

The following terms were defined in terms of how they work and how they are understood as follows:

Appearance. Refers to how the cookies made with jackfruit seed flour look in terms of color, shape, and overall presentation.

Aroma. Refers to the smell of the jackfruit seed flour cookies that can affect how people find them appealing.

Consumer Acceptance. Refers to how much the consumers like or approve of the jackfruit seed flour cookies after tasting and evaluating them.

Cookies. Refers to baked snacks made by partly replacing wheat flour with jackfruit seed flour.

Flavor. Refers to the taste and smell experienced when eating the jackfruit seed flour cookies.

Food Preparation Method. Refers to the steps used in making jackfruit seed flour, including washing, drying, roasting, grinding, and sifting the seeds.

Health Benefits. Refers to the advantages a person may get from eating cookies made with jackfruit seed flour, such as added nutrients and natural compounds that help the body.

Jackfruit Seed Flour (JSF). Refers to flour made by drying and grinding jackfruit seeds, used to replace part of the wheat flour in baking.

Nutritional Content. Refers to the nutrients found in the jackfruit seed flour cookies, such as protein, fiber, carbohydrates, vitamins, and minerals.

Sensory Qualities. Refers to the qualities of the jackfruit seed flour cookies that can be experienced through the senses, including how they taste, smell, feel, and look.

Shelf Life. Refers to how long the jackfruit seed flour or the cookies made from it can stay safe and sound to eat when appropriately kept.

Taste. Refers to the flavour of the jackfruit seed flour cookies, including sweetness and other sensations felt when eating them.

Texture. Refers to how the cookies feel when eaten, such as being soft, chewy, or crunchy, depending on the amount of jackfruit seed flour used.

CHAPTER II

RESEARCH METHODOLOGY

This chapter will provide more detail on the research methods and procedures. It will discuss the research design, the study setting, the participants, ethical issues, the interview guide, and the data collection process.

2.1 Research Design

The research method that will be employed in this study will be an experiment, as is common in areas such as hospitality, science, finance, and education. It will also be helpful to businesses and organizations for testing novel approaches and products. The phenomenon that will be explored in the experiment will be how jackfruit seed flour will be used in the development of the cookie product as a delicacy in Nasipit. It will be done to boost food security through alternative sources of food. Additionally, more efficient agricultural systems, including eco-friendly practices, will promote biodiversity, healthy soil, and sustainable agriculture.

2.2 Research Materials and Methods

The main research instrument that will be used in this experimental study will be a self-made sensory evaluation and survey questionnaire. The questionnaire will be designed to assess the acceptability and quality attributes of the developed cookie product using jackfruit seed flour as a substitute for wheat flour. The variables and criteria for evaluation will be anchored on the study by Cagula et al. (2025) among Saint Michael College of Caraga students titled: *"Product Innovation: The Impact of Malunggay Integration to Chicharon Development Food."* Before the data collection, the instrument will undergo pilot testing to ensure its reliability and validity.

The research instrument will consist of three sections. Part I will present the demographic profile of the respondents. This part will collect basic information such as their age and which school they belong to. Part II will focus on the sensory evaluation of the developed cookie, assessing attributes

such as appearance, aroma, texture, taste, and overall acceptability using a Likert scale: 4 for Strongly Agree, 3 for Agree, 2 for Disagree, and 1 for Strongly Disagree. Part III will measure the respondents' perception of the product's appearance, aroma, flavour, and texture. The purpose of this instrument will be to gather reliable feedback that will guide improvements in the formulation and will contribute to the development of a nutritious and marketable cookie product using jackfruit seed flour.

Statistical Treatment

In presenting, analyzing, and interpreting data, the researcher will use the following statistical test, namely.

Frequency and Percentage – This will be used to summarize the demographic profile of the respondents.

Weighted Mean - This tool will be used to determine the level of acceptability of the cookie product in terms of its sensory attributes (appearance, aroma, texture, taste, and overall acceptability).

Spearman rho Correlation – This will be applied to determine the relationship between the rank sensory preferences of the respondents and their perception of the product's appearance, aroma, texture, taste, and overall acceptability.

2.3 Research Environment

The study will be conducted in Nasipit, Agusan del Norte, a coastal municipality with growing economic activity and accessible educational institutions. The area also has abundant agricultural resources, particularly jackfruit, which supports the study's aim of developing locally inspired and sustainable food products. The experimental procedures, including the preparation and baking of the cookies, will be carried out by the researchers in the kitchen laboratory of Saint Michael College of Caraga (SMCC). Established in 1948, SMCC is recognized as one of the pioneer institutions in the Caraga Region. The kitchen laboratory will serve as the main site for producing the cookies using jackfruit seed flour to explore its potential as a natural flavor enhancer. The taste testing and sensory evaluation will not be limited to SMCC. Respondents will come from a selected urban school in Nasipit, Agusan del Norte, allowing the study to gather feedback from a broader group of students. They will evaluate the cookies based on flavor, texture, and overall acceptability. Their responses will help determine the nutritional value, consumer acceptance, and potential of jackfruit seed flour in promoting healthier food options within schools and local communities.

2.3 Research Population

The participants will be selected using stratified random sampling. The number of participants from each school will be determined proportionally based on the total population of Senior High School students in each school. This ensures that schools with larger student populations have more participants while smaller schools have fewer, maintaining fair representation across all schools. Official enrollment data will be obtained through a formal request letter addressed to the respective school administrators. From each school's list of eligible students, participants will be randomly selected within their respective groups to give every

student an equal chance of being chosen. Inclusion criteria are students who are officially enrolled in the Senior High School program and are 18 years and above at the time of data collection. Exclusion criteria are students who are below 18 years old, those not officially enrolled in the Senior High School program, and students with health-related conditions such as allergies or other medical issues that may pose risks.Limiting participation to students who are 18 years and above ensures ethical compliance and protects minors in accordance with established research standards of the U.S. Department of Health and Human Services.

2.3 Table 1. Research Sample

CONSUMERS	SAMPLE	
	F	%
Calixto D. Montinola National High School (CDMNHS)	112	14.3
Nasipit National Vocational School (NNVS)	307	39.1
Northwestern Agusan Colleges (NAC)	70	8.9
Saint Michael College of Caraga (SMCC)	296	37.7
Total	785	100.0

Table 1 shows the distribution of the respondents of the study, consisting of Senior high school students from the selected schools.

2.4 Research Data and Analysis Plan

The research will be conducted in selected schools in Nasipit, Agusan del Norte that offer Senior High School programs. Before the research begins, request letters will be formally sent to the school heads, as well as to the Registrar of Saint Michael College of Caraga, requesting permission to pursue the research. The study will be conducted in a series of processes, including preparation, experimentation, and data interpretation. The researchers are the students taking the course of Hospitality Management of Saint Michael College of Caraga, who have basic knowledge of food preparation and food tasting.

Data will be gathered through an experimental method. The selection of respondents will be through convenience sampling, where participants will be chosen based on availability and willingness. The experiment will be conducted in the kitchen laboratory of Saint Michael College of Caraga located on the main campus. At this stage, the researchers will be preparing cookies with added jackfruit seed flour and testing the possibility of its use as a natural flavor enhancer. The purpose of the study and the ingredients used in making the jackfruit seed cookies will be explained to the respondents before the actual taste test. The packaging will indicate a complete list of ingredients so that the respondents would know what is contained in the product. Respondents will be asked if they have food allergies, and discomfort or refusal to participate will not require them to join the taste test.

The rights of the respondents include voluntary participation and privacy in answering. These rights will be discussed during the activity. Consent forms will be given and signed indicating that the respondents agree to take part in this research. The respondents can also correct their responses in case they made some mistakes.

To analyze the jackfruit seed cookies, the respondents will be presented with three different samples consisting of varied quantities of jackfruit seed flour. To analyze the samples, a checklist

will be given, and the results will be recorded. The responses for taste, texture, aroma, and overall acceptance will be recorded using a scale, while the questionnaire consisting of prepared questions will also be used for extra comments.

All the data collected will be analyzed using the help of a statistician to make sure the results are accurate and consistent.

Ingredients

Below are the ingredients that will be needed for the integration of jackfruit seeds into cookie development.

Jackfruit Seeds.

These will serve as the main innovative ingredient in the cookie formulation. Ground jackfruit seeds add a nutty flavour and a slightly chewy texture, while enhancing the nutritional value with added protein and fibre.



Figure 1. Jackfruit Seeds.

All-purpose flour. It will present the skeletal foundation of the cookie dough, making it suitable in terms of texture and stability when combined with the other elements.



Figure 2. All-purpose flour.

Granulated Sugar. It will be a sweetener, help with the crispness of cookies, and also help with proper browning.



Figure 3. Granulated Sugar

Brown Sugar. It will increase their humidity, chewiness, and richness of the cookies, adding a slight caramel flavour.



Figure 4. Brown Sugar

Baking soda. It will serve as a raising agent, helping the cookies rise a bit and providing a light, soft texture.



Figure 5. Baking Soda

Salt. It will also offset the sweetness and improve the general flavour profile of the jackfruit seed cookies.



Figure 6. Salt

Lemon Zest. It will bring a cooling scent and a slight hint of citrus that will go well with the nuttiness of jackfruit seeds.



Figure 7. Lemon Zest

Butter. It will make the texture rich and smooth, adding tenderness and flavour to the cookies.



Figure 8. Butter

Vanilla Extract. It will make the cookies smell and add richness to the flavour, making them more aromatic and pleasing.



Figure 9. Vanilla Extract

Egg. It will act as a binding agent, holding the ingredients together, and give the mixture softness by providing moisture.

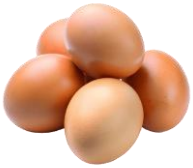


Figure 10. Egg

Chocolate Chips. They will provide sugary, crunchy flavours, making cookies even tastier.



Figure 11. Chocolate Chips

Milk. It will provide a creamy, smooth flavor and help improve the cookies' texture, making them softer and richer.



Figure 12. Milk

Materials

These are the materials that will be needed to complete the study, as listed below.

Tray. It will serve as a base to evenly spread the jackfruit seeds during drying, ensuring proper air circulation for consistent drying.



Figure 13. Tray

Net Cover. It will be used to cover the jackfruit seeds during drying, protecting them from dust, insects, and other contaminants while still allowing airflow.



Figure 14. Net Cover

Knife. It will be used to cut the jackfruit seeds into smaller pieces before drying, making them easier to roast and grind into flour.



Figure 20. Knife

Mixing bowls. It will be one of the necessary tools for making Jackfruit Seeds Cookies, along with the ingredients.



Figure 12. Mixing Bowls

Measuring cups and spoons. It will be an essential apparatus for measuring all ingredients, including jackfruit seed powder.



Figure 13. Measuring cups and spoons

Whisk. This will be a crucial article on combining the dry and moist ingredients, including jackfruit seed powder.



Figure 14. Whisk

Cookie Cutter. It will be applied to cut the dough of cookies into the exact sizes so that they look the same and bake the same.



Figure 21. Cookie Cutter

Spatula. The cookie mixture will be stirred and loosened during the dry-frying process to even out the heat and prevent ingredients from sticking to the pan.



Figure 12. Spatula

Blender. It will be an essential tool for finely grinding jackfruit seeds, ensuring that all dry and moist ingredients blend seamlessly for a consistent texture, even nutrient distribution, and a perfect combination.



Figure 12. Blender

Cookie Sheet Pan. It will be the place where the mixture was put after all was ready. To prepare the Jackfruit seed bake product.



Figure 13: Cookie Sheet Pan

Frying Pan. The jackfruit seeds will be roasted or left to dry to remove excess moisture, then roasted to improve the flavour before grinding.



Figure 18. Frying Pan

Oven. It will be applied during the baking of jackfruit seed cookies, providing the right amount of heat to ensure the end product has the right texture and taste.



Figure 19. Oven

Personal Protective Equipment

Hand Gloves. These will be essential to maintain the cleanliness of the preparation process. They also protected the jackfruit seed ingredients and other baking ingredients from contamination, thereby ensuring the cleanliness, quality, and safety of the cookies.



Figure 16. Hand Gloves

Face Masks. These will play a significant role in ensuring hygiene and preventing contamination of saliva during preparation, ensuring that Jackfruit Seeds Cookies are safe and of high quality.



Figure 17. Face Mask

Chef Hat. It will be defined as putting on a cap to prevent hair in the mixture from ending up in the cookies, keeping the work area clean, and ensuring hygiene throughout the baking process.



Figure 18. Hat

Apron. It will be used to cover the baker and the area where baking is done, preventing contamination and ensuring the cookies remain clean and of high quality.



Figure 19. Apron

Chef Jackets. It will be applied to ensure safety and cleanliness during preparation; it will act as a shield against contamination and help maintain a neat baking environment.



Figure 20. Chef Jackets

Chef Pants. As part of the protective gear, these will provide full leg coverage to prevent any contaminants from coming into contact with the baking ingredients during preparation.



Figure 21. Chef Pants

Clogs. These will be worn as protective footwear to prevent slips and ensure safety in the working area during the baking process.



Figure 22. Chef Clogs

Preparation of Jackfruit (Artocarpus Heterophyllus)

The study will follow the steps in the flowchart plan shown below.

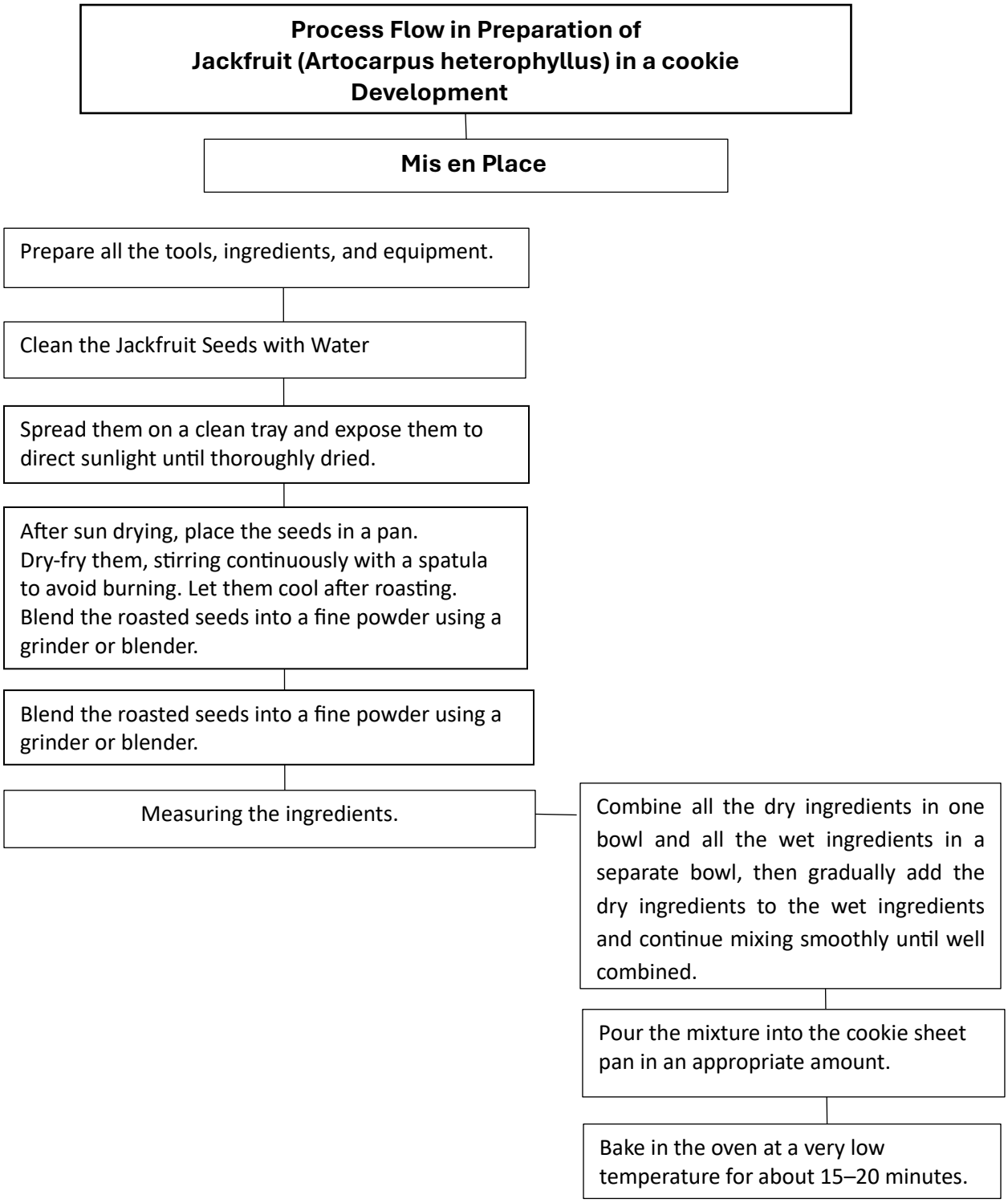


Figure 23. The diagram shows the process flow of Jackfruit Cookies.

Part I. Procedure for Making Jackfruit Cookies. For product A 30% J - 70% AF.

1. Fresh, mature jackfruit seeds will be collected and thoroughly washed to remove any dirt or foreign materials.

2. The cleaned seeds will be boiled for 30 minutes to soften them, then peeled to obtain the white kernel portion.

3. The kernels will be sliced into small pieces and dried in an oven at 60°C until completely dehydrated.

4. The dried jackfruit seed pieces will be ground using a blender to produce fine flour, and a strainer will be used to remove coarse particles. The jackfruit seed flour will be stored in an airtight container for later use.

5. For the cookie mixture, 1/3 cup plus one teaspoon of jackfruit seed flour (40%) and 3/4 cup plus one tablespoon plus one teaspoon of all-purpose flour (60%) will be measured and mixed in a clean bowl.

6. In a separate bowl, 1/4 cup of butter and 1/4 cup of brown sugar will be creamed together until light and soft.

7. One egg and one teaspoon of vanilla extract will be added to the creamed mixture and thoroughly mixed to achieve consistency.

8. The combined flour mixture will be gradually added to the wet ingredients while continuously stirring to form a smooth dough.

9. The dough will be rolled and shaped into small, round portions using a cookie cutter, then will be placed evenly on a baking tray lined with parchment paper.

10. The cookies will be baked in a preheated oven at 180°C for 10–12 minutes or until golden brown.

11. Once baking is completed, the cookies will be allowed to cool for 5 minutes at room temperature before being stored in airtight containers for evaluation and consumption.

Part I. Procedure for Making Jackfruit Cookies. For product B 50% J - 50% AF.

1. Fresh, mature jackfruit seeds will be collected and thoroughly washed to remove any dirt or foreign materials.

2. The cleaned seeds will be boiled for 30 minutes to soften them, then peeled to obtain the white kernel portion.

3. The kernels will be sliced into small pieces and dried in an oven at 60°C until completely dehydrated.

4. The dried jackfruit seed pieces will be ground using a blender to produce fine flour, and a strainer will be used to remove coarse particles. The jackfruit seed flour will be stored in an airtight container for later use.

5. For the Jack Fruit Cookie (half batch), 1/2 cup + 1 tablespoon + 1 teaspoon of jackfruit seed flour (50%) and 1/2 cup + 1 tablespoon + 1 teaspoon of all-purpose flour (50%) will be measured and placed in a mixing bowl.

6. One egg and one teaspoon of vanilla extract will be added to the creamed mixture and thoroughly mixed to achieve consistency.

7. The combined flour mixture will be gradually added to the wet ingredients while continuously stirring to form a smooth dough.

8. The dough will be rolled and shaped into small, round portions using a cookie cutter, then will be placed evenly on a baking tray lined with parchment paper.

9. The cookies will be baked in a preheated oven at 180°C for 10–12 minutes or until golden brown.

10. Once baking is completed, the cookies will be allowed to cool for 5 minutes at room temperature before being stored in airtight containers for evaluation and consumption.

Part I. Procedure for Making Jackfruit Cookies. For product C 60% J - 40% AF.

1. Fresh, mature jackfruit seeds will be collected and thoroughly washed to remove any dirt or foreign materials.

2. The cleaned seeds will be boiled for 30 minutes to soften them, then peeled to obtain the white kernel portion.

3. The kernels will be sliced into small pieces and dried in an oven at 60°C until completely dehydrated.

4. The dried jackfruit seed pieces will be ground using a blender to produce fine flour, and a strainer will be used to remove coarse particles. The jackfruit seed flour will be stored in an airtight container for later use.

5. For the Jackfruit Cookie (half batch), the total flour volume of 1.125 cups will be measured. At 60% jackfruit seed flour, 2/3 cup + 1 tablespoon will be taken as jackfruit seed flour, and the remaining 40% (1/2 cup + 1 tablespoon + 1 teaspoon) will be taken as all-purpose flour, then will be mixed in a clean bowl.

6. One egg and one teaspoon of vanilla extract will be added to the creamed mixture and thoroughly mixed to achieve a consistent texture.

7. The combined flour mixture will be gradually added to the wet ingredients while continuously stirring to form a smooth dough.

8. The dough will be rolled and shaped into small, round portions using a cookie cutter, then will be placed evenly on a baking tray lined with parchment paper.

9. The cookies will be baked in a preheated oven at 180°C for 10–12 minutes or until golden brown.

10. Once baking is completed, the cookies will be allowed to cool for 5 minutes at room temperature before being stored in airtight containers for evaluation and consumption.

Part II. Costing and Pricing

Jackfruit Seeds	20 pesos
500 ml All Purpose Flour	25 pesos
113 g Butter	35 pesos
75 g Granulated sugar	10 pesos
75 g Brown sugar	10 pesos
1 large Egg	12 pesos
Vanilla extract	5 pesos
Baking soda	1 pesos
Salt	1 pesos
Lemon Zest	15 pesos
Milk	35 pesos
45 g Chocolate chips	30 pesos
TOTAL	**199**

Statistical Treatment

Statistics is a method of analysis that uses different tools to examine and understand data. It will be used to assess and understand the results of the data gathered in the study. The researchers will use the following statistical tools, as these are suitable for the type of data to be collected.

Frequency and Percentage: This method will be used to analyse the responses for statements 1, 2, and 4 to describe the respondents' profiles. It will show how often each response appears and its percentage of the total.

Mean: This method will be used to analyse statement number 3. It will determine the sensory qualities and acceptability of baked products made with jackfruit seed waste through sensory evaluation.

2.5 Ethical Standards

A. Protection of Intellectual Property Rights (IPR)

This study will focus on new products. No new technologies or processes requiring patenting will be developed. The survey and reports are original works protected by copyright to preserve the integrity of the research. No proprietary materials are used, and such public resources and academic references are examined in accordance with copyright laws so that no trademarks will be violated.

B. Informed Consent

The researchers will explain the study's purpose, why the respondents will be selected, and the possible benefits and risks of participating in the survey. Respondents will be able to withdraw anytime, ask questions, or skip any questions they find uncomfortable. Before conducting the study, the researchers will send formal request letters to the selected schools. The researchers will ask respondents about any known allergies in advance to ensure safety during all sensory tests. The cookies will be packaged in sanitary and hygienic packaging to prevent contamination and ensure suitability for tasting. The information, including the ingredients and allergy-related details, will be visible on the packaging. All personal data will be collected securely and kept private. When applicable, respondents will be offered a printout of the survey transcripts for review and will be allowed to change, delete, or add answers. Feedback will be requested later to validate and improve the study's results. Respondents will sign a waiver to confirm their participation. Finally, the respondents will be fully informed of all ingredients used in the cookies before the sensory test to prevent any harm or adverse reactions.

C. Data Privacy and Confidentiality

Personal information is collected through encrypted forms or surveys and anonymized to protect respondents' privacy. Information will be stored in password-protected files, with access restricted to authorized team members. Wherever data will be disseminated to external parties, respondents have provided express consent.

D. Voluntary Participation and Freedom to Withdraw

Study participation is voluntary for all respondents, and they are advised that they may withdraw at any time without penalty. Ethical objections to the use of animals are irrelevant because no animals are involved in the study. Human research ethical standards are adhered to throughout.

E. Minimization of Harm and Risk Management

To minimize the burden on respondents, the study will ensure that participation is voluntary and that any risks are clearly explained. The respondents are not forced to try the products, and the researchers will ask them whether they are allergic to them before they join the research. Safe procedures will be used to prevent harm from the questionnaire, survey, and sensory test. Proper packaging will be used to ensure the product is secure, protected against contamination, and tastes appropriate. The respondents are told they can withdraw at any time if they are in distress. The school clinic will be available if a respondent has an allergic reaction during the taste test. It does not apply to the 3Rs (Replacement, Reduction, Refinement) because no animals will be harmed.

F. Beneficence and Contribution to Knowledge

The study will be relevant to the research field, as it will examine new strategies to add value to food products, particularly the use of jackfruit seed flour in place of wheat flour in the production of jackfruit seed cookies. It will encourage food innovations and nutritional progress

by showing how jackfruit seed Higaonon-inspired cookie products, nutrient-dense ingredients, can be turned into a delicacy in Nasipit. Furthermore, the research will maintain transparency, informing respondents of their right to receive findings, fostering trust and ethical engagement.

G. Justice and Fair Participant Selection

The selection criteria for choosing respondents will be open and equitable to ensure a representative sample. There will be no discrimination against people with diverse backgrounds, and no use of animals in the study.

H. Data Integrity and Accuracy

The study will use standard approaches to ensure accuracy and reliability in data collection and analysis. Any errors, limitations, or biases will be documented to preserve the transparency and integrity of the study. Any AI tools used will gather survey data, organize survey responses, and assist with research analysis. They will be thoroughly examined to ensure there are no potential biases in the review of the findings. It will reinforce that there will be no errors during the random data collection in the study.

I. Transparency and Honesty in Reporting

The results of the study will be reported honestly and accurately. Any copyright and patent contents that have been used in the study shall be appropriately credited and referenced as per the academic requirements. The authors state that the research has no conflicts of interest (COIs), which could affect the research findings or conclusions. Moreover, the research is enhanced with different AI tools to improve its quality and efficiency.

J. Use of Patented or Copyrighted Materials

The research does not employ patented technologies, so no approvals from patent holders are required. All other copyrighted material, such as software or literature use, has proper clearance and was appropriately referenced.

K. Ethical Considerations for Animal and Human Trials

Respondents participate in the sensory assessment after being fully informed of the possible risks and benefits to them as subjects, to enable informed consent. The respondents are fully informed of the study, its objectives, procedures, and the potential data discomfort that may occur. The researchers ensure that the respondents will receive the support services required for their well-being. As no animals were used in the study, ethical clearance for animals is not required.

L. Responsible Use of AI and Other Related Technologies

Digital technologies will support every stage of the research. Elicit AI will quickly identify valuable and reliable research documents, making it easier to access high-quality sources in less time. Anara.com will speed up the literature review by efficiently screening references. This will ensure that credible, verified research is reviewed. Google Scholar will be used to access peer-reviewed, academically credible articles. All scientific sources will therefore be reliable and trustworthy for the research purpose. During the writing stage, ChatGPT will be used to develop ideas, wording, and content, enhancing clarity and coherence in the research narrative. The paper will further improve its professionalism by using Grammarly to correct grammatical, structural, and syntactic errors, ensuring accuracy and readability. All these AI tools will be used responsibly to ensure the research remains valid, reliable, and maintains overall integrity.

M. Ethical Clearance and Institutional Approval

The SMCC Ethical Review Board will provide ethical approval for the study to ensure the research follows institutional standards for human participants.

Research Instrument

Saint Michael College of Caraga
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UTILIZING JACKFRUIT SEED WASTE IN HEALTHIER BAKED PRODUCTS: A HIGAONON-INSPIRED PRODUCT DEVELOPMENT

PART I. Demographic Profile. This part of the questionnaire will focus on establishing the respondents' demographic profile by collecting information on their age, grade level, and place of residence.

Instructions: Please put a check mark (✓) in the box that corresponds to your answer.

(Palihug markahi ug tsek sa lingin nga nahiangay sa imong tubag.)

A. Age

☐ 18-20 years old

☐ Years old and above

B. Type of Respondents

☐ **CDMNHS Student**
(Estudyante sa CDMNHS)

☐ **NAC Student**
(Estudyante NAC)

☐ **NNVS Student**
(Estudyante sa NNVS)

☐ **SMCC Student**
(Estudyante sa SMCC)

PART II.

Directions: Using the rating scale below

Please put a Check (√) inside the box that indicates your level of agreement for each statement.

Scale	Verbal Description	Verbal Interpretation
4	**Strongly Agree (SA)** (*Hugot na Pag-uyon)*	**-The quality is highly acceptable.** *(Ang kalidad kay taas kaayo nga pagdawat.)*
3	**Agree (A)** *(Uyon)*	**-The quality is moderately acceptable.** *(Ang kalidad kay medyo madawat.)*
2	**Disagree (D)** *(Dili Uyon)*	**-The quality is slightly acceptable.** *(Ang kalidad kay dili kaayo dawat.)*
1	**Strongly Disagree (SD)** *(Hugot na Dili Pag-Uyon)*	**-The quality is not acceptable.** *(Ang kalidad kay dili madawat.)*

Product A. **30% Jackfruit Seeds 70% All Purpose Flour**	**Strongly Agree** **SA** **(4)**	**Agree** **A** **(3)**	**Disagree** **D** **(2)**	**Strongly Disagree** **SD** **(1)**
APPEARANCE **The Jackfruit Seeds cookie has visual appeal.** *(Ang itsura sa Jackfruit Seeds cookie makapadani sa mata.)*				
AROMA **The Jackfruit Seeds offer a distinct scent.** *(Ang Jackfruit Seeds naghatag og lahi nga baho.)*				
FLAVOR **Jackfruit Seeds are nutritious but have a bitter taste.** *(Jackfruit Seeds, masustansya apan pait nga lami.)*				
The flavor of Jackfruit Seeds is easily detectable. *(Dali mamatikdan ang lasa sa Jackfruit Seed)*				
TEXTURE **The Jackfruit Seeds offers a crunchiness texture in the mouth.** *(Mabatian ang ka gumkom sa Jackfruit Seeds ngadto sa baba.)*				
Comment/s: *(Kumento)*				

Product B. **50% Jackfruit Seeds 50% All Purpose Flour**	**Strongly Agree** **SA** **(4)**	**Agree** **A** **(3)**	**Disagree** **D** **(2)**	**Strongly Disagree** **SD** **(1)**
APPEARANCE **The Jackfruit Seeds cookie has visual appeal.** *(Ang itsura sa Jackfruit Seeds cookie makapadani sa mata.)*				
AROMA **The Jackfruit Seeds offer a distinct scent.** *(Ang Jackfruit Seeds naghatag og lahi nga baho.)*				
FLAVOR **Jackfruit Seeds are nutritious but have a bitter taste.** *(Jackfruit Seeds, masustansya apan pait nga lami.)*				
The flavor of Jackfruit Seeds is easily detectable. *(Dali mamatikdan ang lasa sa Jackfruit Seed)*				
TEXTURE **The Jackfruit Seeds offers a crunchiness texture in the mouth.** *(Mabatian ang ka gumkom sa Jackfruit Seeds ngadto sa baba.)*				
Comment/s: *(Kumento)*				

Product C. **60% Jackfruit Seeds 40% All Purpose Flour**	**Strongly Agree** **SA (4)**	**Agree** **A (3)**	**Disagree** **D (2)**	**Strongly Disagree** **SD (1)**
APPEARANCE **The Jackfruit Seeds cookie has visual appeal.** *(Ang itsura sa Jackfruit Seeds cookie makapadani sa mata.)*				
AROMA **The Jackfruit Seeds offer a distinct scent.** *(Ang Jackfruit Seeds naghatag og lahi nga baho.)*				
FLAVOR **Jackfruit Seeds are nutritious but have a bitter taste.** *(Jackfruit Seeds, masustansya apan pait nga lami.)*				
The flavor of Jackfruit Seeds is easily detectable. *(Dali mamatikdan ang lasa sa Jackfruit Seed)*				
TEXTURE **The Jackfruit Seeds offers a crunchiness texture in the mouth.** *(Mabatian ang ka gumkom sa Jackfruit Seeds ngadto sa baba.)*				
Comment/s: *(Kumento)*				

The research instrument in this study will be based on the original tool developed by students from Saint Michael College of Caraga, as conducted by Cagula et al. (2025). The questionnaire will be developed based on their tool to be used in this study. Certain items will be adapted to better align with the specific objectives and context of the proposed research, while maintaining the core structure and intent of the original instrument.

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